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Solvable Lie Algebras

1.1 Solvable Lie Algebras

Lemma 1.1.1

Suppose that I is an ideal of L . Then L/I is abelian if and only if I contains the derived algebra L' .

Proof. L/I is abelian, if and only if for each $x, y \in L$,

$$[x + I, y + I] = [x, y] + I = 0 + I \iff [x, y] \in I$$

That is L/I is abelian, if and only if $L' = [L, L] \subseteq I$. □

Definition 1.1.2

This lemma tells us that the derived algebra L' is the smallest ideal of L with an abelian quotient. By the same argument, the derived algebra L' itself has a smallest ideal whose quotient is abelian, namely the derived algebra of L' , which we denote $L^{(2)}$, and so on. We define the **derived series** of L to be the series with terms

$$L^{(1)} = L' \quad \text{and} \quad L^{(k)} = [L^{(k-1)}, L^{(k-1)}] \text{ for } k \geq 2.$$

Then $L \supseteq L^{(1)} \supseteq L^{(2)} \supseteq \dots$. As the product of ideals is an ideal, $L^{(k)}$ is an ideal of L (and not just an ideal of $L^{(k-1)}$).

Definition 1.1.3

The Lie algebra L is said to be **solvable** if for some $m \geq 1$ we have $L^{(m)} = 0$.

Lemma 1.1.4

If L is a Lie algebra with ideals

$$L = I_0 \supseteq I_1 \supseteq \dots \supseteq I_{m-1} \supseteq I_m = 0$$

such that I_{k-1}/I_k is abelian for $1 \leq k \leq m$, then L is solvable.

Note.

- 我们把这样的理想链视作过滤掉非交换性的多层滤网, 每次都把可交换的元 (相当于粗颗粒) 保留, 将不能交换的程度 (细颗粒) 过滤下去.
- 导出列实现对不可交换性的精准过滤和采集.
- 普通的理想链是可能更粗的滤网, 它能承载所有上一层级的非交换性, 但是可能混入可交换性的细颗粒.
- 这个引理相当于是说, 如果我们通过可能更不精细的手法将所有的非交换性过滤干净, 那么原有的精确过滤也一定能过滤干净.

Proof. We shall inductively shows that $L^{(k)}$ is contained in I_k , then by taking $k = m$, we get $L^{(m)} = 0$.

Since $L/I_1 = I_0/I_1$ is abelian, $I_1 \supseteq L'$. From inductive hypothesis, we know that $L^{(k-1)} \subseteq I_{k-1}$, since I_{k-1}/I_k is abelian, that is

$$[I_{k-1}/I_k, I_{k-1}/I_k] = 0$$

which is equivalent to

$$[I_{k-1}, I_{k-1}] \subseteq I_k$$

Since $L^{(k-1)} \subseteq I_{k-1}$, we have

$$L^{(k)} = [L^{(k-1)}]$$

□

Proposition 1.1.5

Suppose that $\varphi : L_1 \rightarrow L_2$ is a surjective homomorphism of Lie algebras. Show that

$$\varphi(L_1^{(k)}) = (L_2)^{(k)}.$$

Proof. For $k = 0$,

$$\varphi(L_1) = L_2$$

Inductively,

$$\varphi(L_1^{(k)}) = \varphi([L_1^{(k-1)}, L_1^{(k-1)}]) = [\varphi(L_1^{(k-1)}), \varphi(L_1^{(k-1)})] = [L_2^{(k-1)}, L_2^{(k-1)}] = L_2^{(k)}$$

□

Lemma 1.1.6

Let L be a Lie algebra.

- (a) If L is solvable, then every subalgebra and every homomorphic image of L are solvable.
- (b) Suppose that L has an ideal I such that I and L/I are solvable. Then L is solvable.
- (c) If I and J are solvable ideals of L , then $I + J$ is a solvable ideal of L .

(a) Omitted.

Note.

- (b) 商去 I 是可解的, 当且仅当各层级的非交换性可以被 I 给兜住.

From the above proposition, we have $(L/I)^{(k)} = (L^{(k)} + I)/I$. If L/I is solvable, then $L^{(m)} \subseteq I$ for some m . We have for some k ,

$$(L^{(m)})^{(k)} \subseteq I^{(k)} = 0$$

Alternatively, it follows from (a) that $L^{(m)}$ is solvable.

$$L^{(m+k)} = (L^{(m)})^{(k)} = 0$$

- (c) $(I + J)/I \simeq J/I \cap J$ is solvable by (a). And since I is solvable, $I + J$ is solvable by (b).